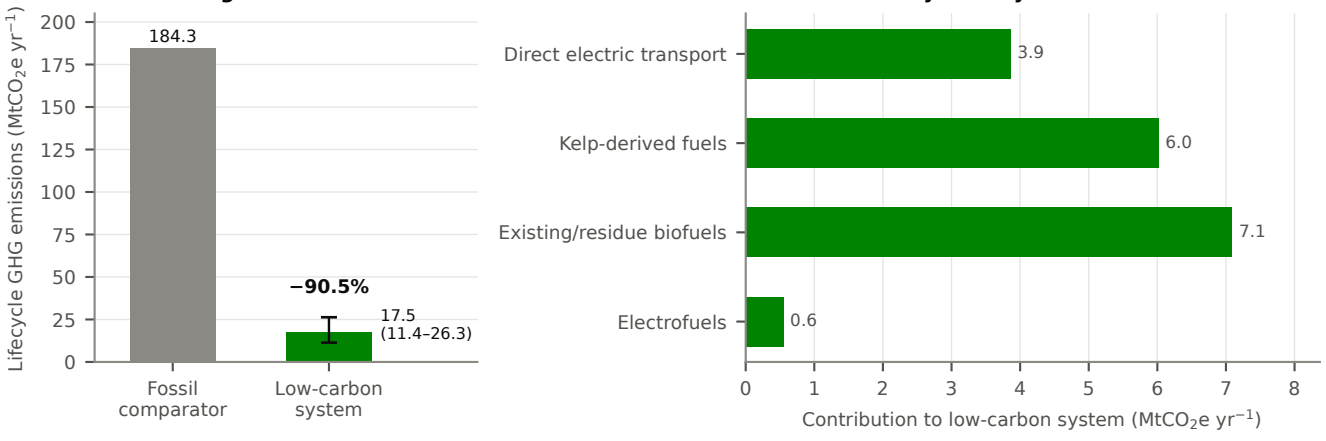
a Screening well-to-wheel emissions
Lifecycle GHG emissions (MtCO2e yr−1)
200
175
150
125
100
75
50
25
0
184.3
−90.5%
17.5
(11.4–26.3)
Fossil comparator
Low-carbon system
b Low-carbon system by carrier
Direct electric transport
Kelp-derived fuels
Existing/residue biofuels
Electrofuels
3.9
6.0
7.1
0.6
0
1
2
3
4
5
6
7
8
Contribution to low-carbon system (MtCO2e yr−1)